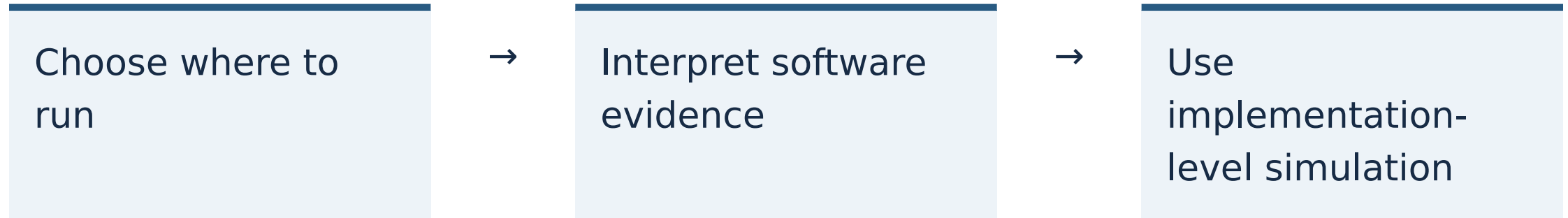


System execution and silicon

Every technique must discharge a named verification claim.



Learning outcomes

Demonstrate these decisions on a real plan row.

- Choose an execution platform with a usable oracle.
- Resolve model disagreements without default authority.
- Interpret GLS and power-aware results within their scope.
- Turn a silicon escape into upstream evidence.

Route through the lesson

Four blocks, each ending with a decision.

- Choose where to run
- Interpret software evidence
- Use implementation-level simulation
- Bring silicon evidence upstream

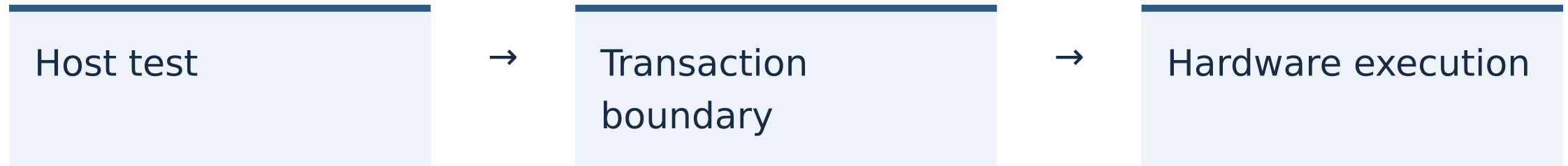
More cycles answer only some questions

Select the platform from the scenario and its oracle.

- Distinguish many short runs from one long execution.
- Budget observability as well as speed.

The host boundary can dominate

Transaction batching matters when work crosses platforms.



Capacity is a shared project resource

Image preparation and debug access belong in the schedule.

- Include compile, queue and turnaround time.
- Allocate platform capacity by verification value.

Discussion: accelerate the wrong workload

Name the evidence that would settle the decision.

- A fast platform spends most time waiting for host calls.
- The scenario itself needs few cycles.
- What boundary or platform decision should be revisited?

Prototypes trade visibility for execution

A long run needs a reproducible failure record.

- Plan observability before the failure arrives.
- Keep software, image and configuration together.

A virtual model has an abstraction boundary

Fast software progress does not validate omitted timing.

- Use the model for the behavior it represents.
- Resolve disagreements against agreed intent.

A hybrid boundary is part of the model

Timing-dependent behavior can invalidate an attractive partition.

- Review synchronization and interaction frequency.
- Retain system checking beyond one driver trajectory.

Discussion: who owns the mismatch?

Name the evidence that would settle the decision.

- A virtual peripheral and RTL disagree.
- Each team declares its artifact authoritative.
- What decision should triage produce?

GLS samples implementation behavior

Static timing analysis remains the timing authority.

- Select runs for implementation-specific risks.
- Keep timing annotation and configuration reproducible.

Unknowns can make a checker silent

Inspect the comparison semantics before accepting a clean run.

- An unknown comparison can bypass a failure branch.
- Check initialization, models and the observer.

Power intent adds temporal obligations

Isolation and restoration must protect the observation boundary.

- Check corruption, isolation and state restoration.
- Exercise the selected power transitions.

Discussion: clean output during power-off

Name the evidence that would settle the decision.

- A powered-down block remains readable in the model.
- Its outputs stay legal throughout shutdown.
- What modeling premise must be challenged?

Trace storage fixes a tradeoff

More observed signals leave fewer cycles in a fixed buffer.



Bring-up is an evidence ladder

A failed step is interpretable only after its prerequisites hold.

- Establish clocks, reset and access first.
- Advance through checks with explicit dependencies.

An escape owes an upstream change

A workaround alone leaves the verification gap open.

- Localize by configuration and time.
- Create a pre-silicon reproducer where feasible.
- Update the plan, model or checker that missed it.

Discussion: a successful workaround

Name the evidence that would settle the decision.

- Firmware avoids the failing operating mode.
- The product now survives the customer scenario.
- Which verification work remains?

Exercise: resolve a model disagreement

Choose evidence before choosing an owner.

- A virtual peripheral and RTL disagree on a register side effect.
- State the possible triage outcomes.
- Explain what a shared description fixes and what it cannot fix.

What to change on Monday

Make one existing verification decision reviewable.

- Record the boundary of one virtual model.
- Audit an unknown-sensitive checker.
- Trace a workaround to its upstream verification change.

Scope and next reading

Use the handout for the detailed artifacts and sources.

- Read the System execution and silicon handout and book Chapters 17–20.
- Platform setup recipes and full power-intent syntax remain in the handout.